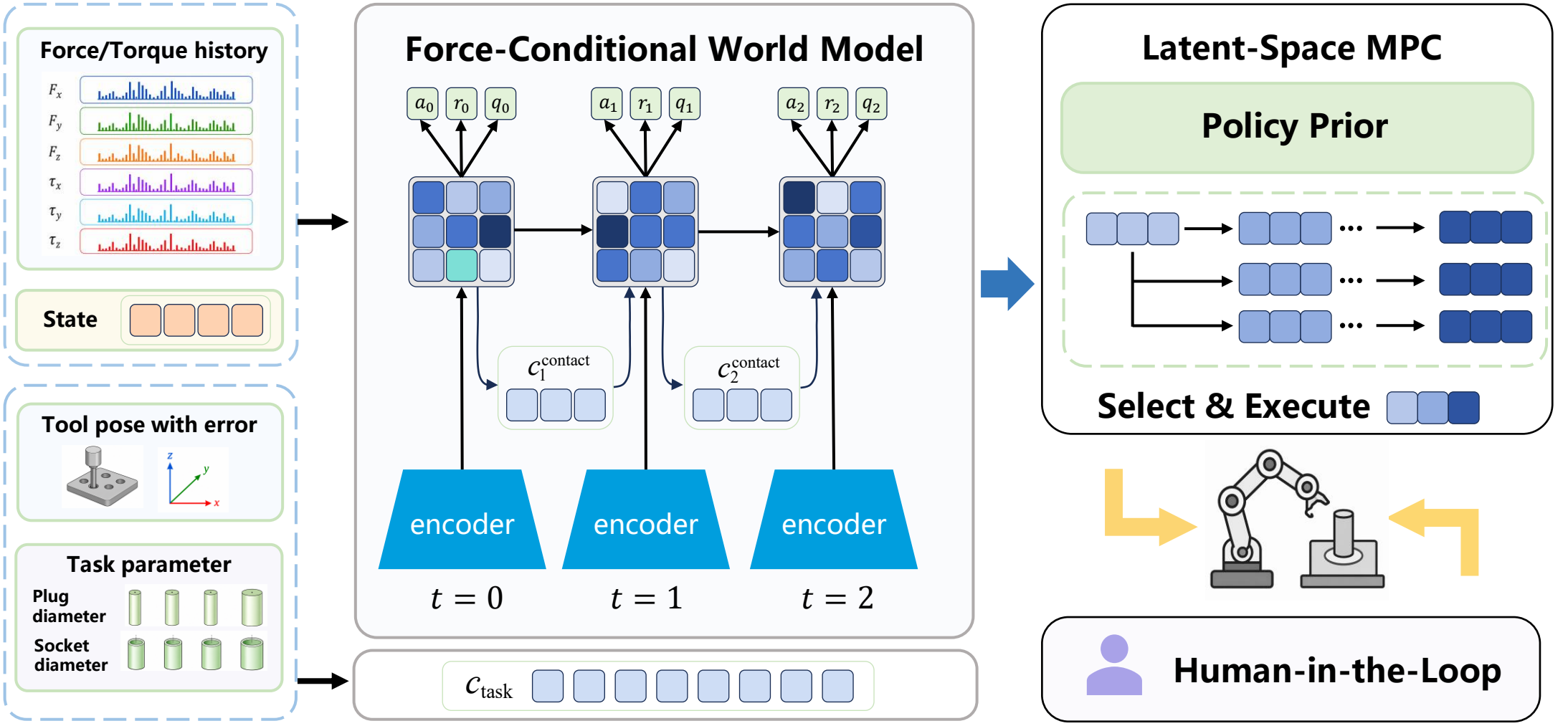

Force/Torque history
F_x
F_y
F_z
τ_x
τ_y
τ_z
State
Tool pose with error
z
y
x
Task parameter
Plug diameter
Socket diameter
Force-Conditional World Model
a_0 r_0 q_0
a_1 r_1 q_1
a_2 r_2 q_2
c_1^{contact}
c_2^{contact}
encoder
encoder
encoder
$t = 0$
$t = 1$
$t = 2$
c_{task}
Latent-Space MPC
Policy Prior
Select & Execute
Human-in-the-Loop